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Sampling, Distribution, Dispersal

Species Diversity, Abundance, and Host Preferences of Mosquitoes (Diptera: Culicidae) in Two Different Ecotypes of Madagascar With Recent RVFV Transmission

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ABSTRACT Mosquito diversity and abundance were examined in six Madagascan villages in either arid (Toliary II district) or humid (Mampikony district) ecotypes, each with a history of Rift Valley fever virus transmission. Centers for Disease Control and Prevention light traps without CO_2 (LT) placed near ruminant parks and animal-baited net trap (NT) baited with either zebu or sheep/goat were used to sample mosquitoes, on two occasions between March 2011 and October 2011. *Culex tritaeniorhynchus* (Giles) was the most abundant species, followed by *Culex antennatus* (Becker) and *Anopheles squamosus/cydippis* (Theobald/de Meillon). These three species comprised more than half of all mosquitoes collected. The NT captured more mosquitoes in diversity and in abundance than the LT, and also caught more individuals of each species, except for *An. squamosus/cydippis*. Highest diversity and abundance were observed in the humid and warm district of Mampikony. No host preference was highlighted, except for *Cx. tritaeniorhynchus* presenting a blood preference for zebu baits. The description of species diversity, abundance, and host preference described herein can inform the development of control measures to reduce the risk of mosquito-borne diseases in Madagascar.

KEY WORDS mosquito, host preference, diversity, trap, Madagascar

Mosquitoes are major vectors of human and animal disease, with economic impacts in endemic and epidemic regions (Morens et al. 2004). In Madagascar, the main mosquito-borne public health concern is malaria, with *Anopheles funestus* (Giles), *Anopheles mascarensis* (de Meillon), and the species complex *Anopheles gambiae* (Giles) as the major vectors (Chauvet et al. 1964, Le Goff et al. 2003, Pock Tsy et al. 2003, Marrama et al. 2004). Arboviruses Rift Valley fever (RVF), West Nile, dengue, and chikungunya have been circulating on the island. RVF virus (RVFV) was first detected in 1989 during the rainy season (Fontenille 1989). The last outbreak of RVFV was reported in 2008 and 2009 during the rainy season (Andriamandimby et al. 2010), in which three mosquito species *Anopheles coustani* Giles, *Anopheles squamosus/cydippis* (Theobald/de Meillon), and *Culex antennatus* (Becker) were involved (Ratovonjato et al. 2011). But, other mosquito species present on the island can also act as potential vectors for RVFV (Balenghien et al. 2013, Tantely et al. 2015). West Nile virus antibodies were first detected on the island in 1978 (Mathiot et al. 1984). Wild cycle of this virus involved several Culicidae species, mainly *Culex*, *Aedes*, and *Anopheles* genera. Fourteen mosquito species were involved in the transmission of this virus in Madagascar (Fontenille 1989, Tantely et al. unpublished). Outbreaks of dengue and chikungunya were also reported in the Eastern coast of Madagascar in 2006 (Ratsitorahina et al. 2008). The two main mosquito species vectors *Aedes albopictus* (Skuse) and *Aedes aegypti* (L.) are present in different areas of Madagascar, but their abundance varies according to the species ecology (Raharimalala et al. 2012).

In Madagascar, studies on mosquito vectors are usually carried out during or after an arbovirus epidemic, allowing implication of responsible vectors through isolation of causative pathogens. For predicting the outbreak of pathogens in one region, it is not sufficient to characterize the risk of pathogen introduction, but it is necessary to inventory the vector species in that region. Behavioral study of vectors, such as host preference, allows determining the probability of vector–host contact. The estimation of species diversity, abundance, and distribution in a time and space scale can be done by sampling adult stage with traps (Brown et al. 2008).

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But, what is the best trap for studying species diversity and species host preference? Several trapping devices are available for monitoring adult mosquitoes, but the performance of each depends on the target species. Few studies consider the behavior and diversity of mosquito species in Madagascar, despite the circulation of pathogens on the island (Grjebine 1966, Tantely et al. 2013).

This study was conducted within the AnimalRisk-OI project (Cardinale et al. 2011), which has the aim to protect populations and livestock from enzootic and epizootic diseases. The study sites were selected using the new distribution map of RVF in Madagascar (Jeanmaire et al. 2011). The districts of Mampikony and Toliary II were chosen for study owing to the difference in their livestock prevalence of RVFV immunoglobulin M (IgM) and immunoglobulin G (IgG). Indeed, in the Mampikony district, IgG rate varied between 40 and 53% and IgM between 0.40 and 0.80%. While, in the Toliary II district, IgG had a rate of 5 to 10% and IgM a rate of 0 to 0.20% (Jeanmaire et al. 2011). Density of livestock in these areas is more than 18 cattle/km^2 (Ministry of Livestock of Madagascar 2007). Moreover, these two regions have ecologically contrasted conditions.

The aim of this study was to determine the mosquito species diversity, abundance, and host preference in locations with a history of RVFV transmission, and assess the efficiency of animal-baited net traps to determine host preference.

Materials and Methods

Study Site. As described above, Mampikony district and Toliary II district were selected for carrying out this study. In the Mampikony district, three sites were sampled: Bekoratsaka (16° 06′24″ S, 47° 29′38″ E), Mampikony I (16° 05′02″ S, 47° 38′25″ E), and Ambalafety (16° 02′43″ S, 47° 38′44″ E; Fig. 1). The district is an ecozone with a wet plain and a long dry season (UNEP/WCMC cited by Jeanmaire et al. 2011). Two distinct seasons exist: the rainy and the dry seasons. Monthly average rainfall varies from 111.1 mm to 368 mm from December to March and from 1.7 mm to 63.3 mm from April to November. Average temperature is 26°C (Centre National de Service Météorologique 1950–2000). In the Toliary II district, sampling was conducted in three sites: Ankilibe (23° 23′52″ S, 43° 44′31″ E), Ankililoaka (22° 43′04" S, 43° 37′28″ E), and Andranovory (22° 08′10″ S, 44° 16′17″ E; Fig. 1). The district is an ecozone with dry or arid plains (UNEP/WCMC cited by Jeanmaire et al. 2011). Two distinct seasons exist: rainy and dry. The rainfall varies from 16 mm to 181.6 mm and heavy rains are observed in January and February. Average temperature is 24°C (Centre National de Service Météorologique 2011). Rainy season in the two districts is from December to May and the dry season is from June to November. In Madagascar, the rainy season typically ends in April, but potential breeding sites can remain flooded until June.

Mosquito Sampling. Entomological investigations were conducted once during the dry season and once during the rainy season. The sampling was conducted in March 2011 for the rainy season and in October 2011 for the dry season in the Toliary II district and in April 2011 for the rainy season and September 2011 for the dry season in the Mampikony district.

The main domestic animals in these regions are zebu (*Bos indicus* (L.)), goat (*Capra hircus* (L.)), and sheep (*Ovis aries* (L.)). In these districts, goat and sheep are raised together. Overnight, these domestic animals are entered in their respective parks located inside the village. The attractiveness of these animals to mosquitoes was studied using animal-baited net traps (NTs; Ba et al. 2006). The NT was baited with either zebu or sheep/goat and was placed close to parks. Knowing the relationship between the relative weights (or surface areas) of the hosts and the proportion of mosquito feeds on each (Port et al. 1980) and to avoid a bias owing to transpiration differences, we used a small zebu comparable with sheep and goat. Each night, four NTs were used, two with zebu, one with goat, and one with sheep, and the comparison of the attractiveness of the bait was made between zebu and goat plus sheep. All traps were georeferenced and ran from 1700 hours (before the sunset) to 0600 hours (following sunrise) over two successive nights. Mosquitoes were collected at 0600 hours using oral aspirators (WHO ref : WHO.USM/00.2a. 2001). Five Centers for Disease Control and Prevention (CDC) light traps without CO_2 (LTs) were placed near or next to livestock over two consecutive nights on the same night as the NTs. The distance from one trap to another was more than 50 m.

Species Determination. Mosquitoes were identified in the field according to morphological taxonomic keys (Grjebine 1966, Fontenille 1989). All the data were consigned with respect to date, locality, genus, species, and method of collection.

Polymerase chain reaction tests were used on a sample belonging to the *An. gambiae* complex to identify *Anopheles arabiensis* (Patton), *Anopheles gambiae ss* (Giles), and *Anopheles merus* (Dönitz; Pock Tsy et al. 2003).

Data Analyses. Beta diversity was estimated by Jaccard's index similarity (J; Jaccard 1908). The study of alpha diversity was conducted by calculating the Simpson's diversity index (D; Simpson 1949) and Shannon's equitability index (E_H; Shannon and Weaver 1949).

The effects of season, district, trap, sites, and species factors on the total number of trapped mosquitoes were tested using an analysis of variance. The effect of trap was tested for four species, *An. coustani, An. squamosus/cydippis*, *Cx. antennatus*, and *Cx. tritaeniorhynchus*, based on their abundance in sampling collection and their status as a potential RVFV vector in Madagascar. For these species, if the NT captured more individuals than the LT significantly, the attractiveness of the bait was tested (i.e., the comparison was between zebu and sheep plus goat). These statistical tests were made using R Core Team (2013).

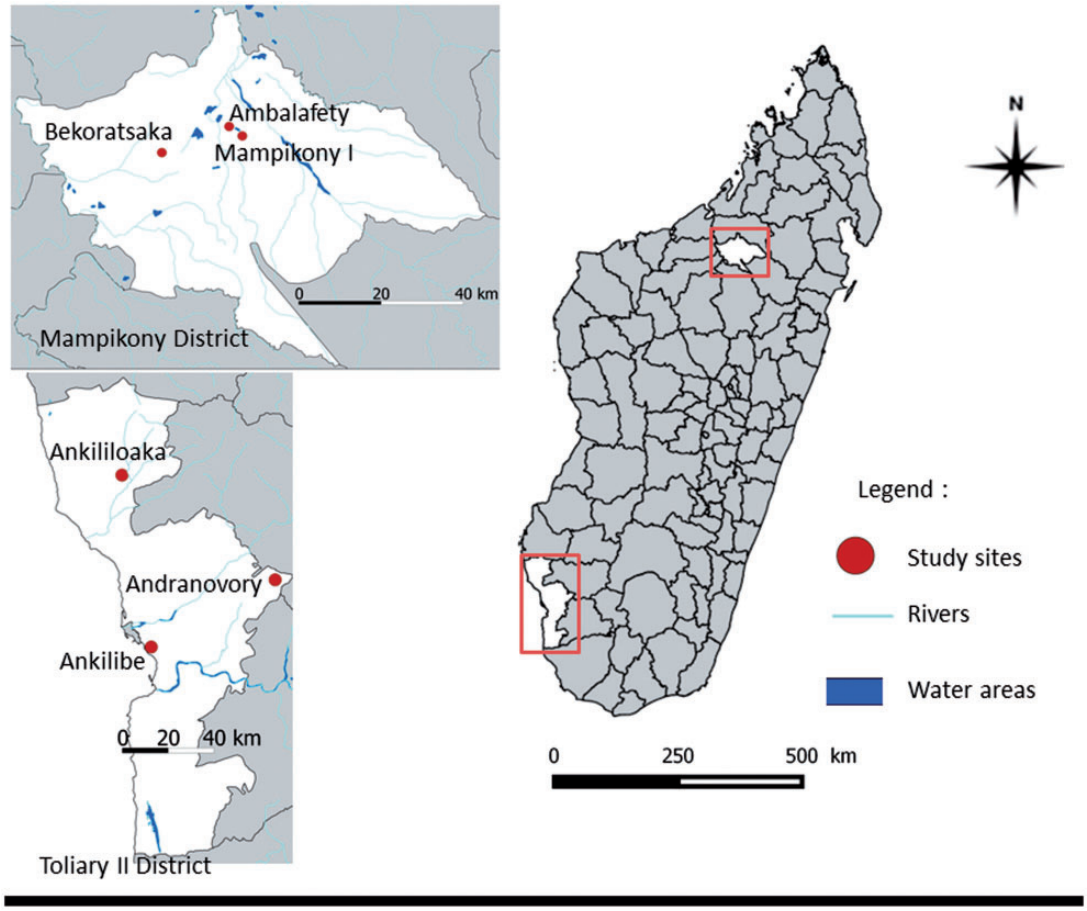


Fig. 1. Study site.

Results

Abundance of Species. In total, 216 trap nights were carried out, i.e., 48 with NT and 60 with LT for each district. These two types of traps collected 11,459 mosquitoes belonging to 35 species attributed to six genera (Table 1). The genus *Culex* comprising 10 species included 6,992 individuals, representing 62.02% of trapped mosquitoes. Fifteen *Anopheles* species (3,322 individuals; 28.99%) and six *Aedes* species (679 individuals; 5.93%) were also caught. The other genera, *Mansonia*, *Aedeomyia,* and *Uranotaenia*, represented 4.07% of the individuals. *Cx. tritaeniorhynchus* was the most abundant species (4,467 mosquitoes; 38.98 %), followed by *Cx. antennatus* (2,111; 18.42%), *An. squamosus/cydippis* (1,127; 9.84%), *Aedes durbanensis* (Theobald; 661; 5.77%), *Anophles pauliani/radama* (Grjebine/de Meillon; 629; 5.77%), *An. coustani* (440; 3.84%), and *Anopheles grassei* (Grjebine; 418; 3.69%; Table1).

In the district of Mampikony, 7,053 (61.55%) mosquitoes with 6 genera and 28 species were captured. The most abundant species are *Cx. tritaeniorhynchus*, *Cx. antennatus*, *An. squamosus/cydippis*, *An. pauliani/radama*, *An. grassei*, and *An. coustani* (Table 1). In the Toliary II district, 4,406 (38.45%) mosquitoes with four genera, and 24 species were collected. The abundant species were formed by *Cx. tritaeniorhynchus* and *Cx. antennatus*. *Ae. durbanensis* was abundant only in the site of Ankilibe (Table 1).

The molecular analyses showed that two members of the *An. gambiae* complex were present in Toliary II: *An. arabiensis* and *An. merus* (Table 1), and two were present in Mampikony: *An. arabiensis* and *An. gambiae* (Table 1).

Trap Effects. In each district, the NTs were more efficient than the LTs ($F = 5.47$, $df = 1$, $P < 0.01$ for the Mampikony district and $F = 9.55$, $df = 1$, $P < 0.01$ for the Toliary II district). The NT collected significantly more mosquitoes, with an average of 234 ± 17 and 262 ± 51 mosquitoes per night per trap, respectively, in the Mampikony district and in the Toliary II district, than the LT, 166 ± 9 and 11 ± 5, respectively.

For the two districts together, NT captured 7,926 (69.17%) mosquitoes, including 6 genera and 33 species, compared with 3,533 (30.83%) mosquitoes caught with the LT, including 5 genera and 29 species. Statistical analyses showed a significant difference among the number of mosquitoes caught by the two traps

Table 1. Total number of each species captured by the two types of trap comprising 216 trap nights: 48 with the animal-baited net trap (NT) and 60 with CDC light traps without CO_2 (LT) for each district

Species	Total number (% total mosquitoes collected)	Mampikony district			Toliary II district		
		Ambalafety	Bekoratsaka	Mampikony I	Andranovory	Ankilibe	Ankililoaka
Aedeomyia furfurea	46 (0.40)	29	1	16	0	0	0
Aedeomyia madagascarica	154 (1.34)	113	10	31	0	0	0
Aedes circumluteolus	4 (0.03)	0	0	4	0	0	0
Aedes durbanensis	661 (5.77)	0	0	0	0	658	3
Aedes fowleri	2 (0.02)	0	0	0	2	0	0
Aedes skusea	10 (0.09)	0	0	0	2	8	0
Aedes sp	1 (0.01)	0	0	0	1	0	0
Aedes tiptoni	1 (0.01)	0	0	0	1	0	0
Anopheles arabiensis	88 (0.77)	7	4	36	24	0	17
Anopheles coustani	440 (3.84)	126	130	141	7	2	34
Anopheles flavicosta	8 (0.07)	0	6	2	0	0	0
Anopheles funestus	8 (0.07)	4	3	1	0	0	0
Anopheles fuscicolor	19 (0.17)	2	0	5	12	0	0
Anopheles gambiae	6 (0.05)	0	6	0	0	0	0
Anopheles grassei	418 (3.65)	114	9	295	0	0	0
Anopheles maculipalpis	65 (0.57)	25	32	6	0	2	0
Anopheles mascarensis	17 (0.15)	1	7	5	4	0	0
Anopheles merus	2 (0.02)	0	0	0	0	2	0
Anopheles pauliani/radama	629 (5.49)	231	45	353	0	0	0
Anopheles pharoensis	254 (2.22)	60	0	193	0	0	1
Anopheles pretoriensis	63 (0.55)	2	47	6	8	0	0
Anopheles rufipes	178 (1.55)	4	70	73	31	0	0
Anopheles squamosus/cydippis	1127 (9.84)	415	217	422	3	0	70
Culex annulioris	2 (0.02)	0	0	0	0	2	0
Culex antennatus	2111 (18.42)	395	186	670	25	19	816
Culex bitaeniorhynchus	29 (0.25)	10	0	16	0	0	3
Culex decens	43 (0.38)	16	8	11	6	2	0
Culex giganteus	7 (0.06)	2	0	0	5	0	0
Culex pipiens	3 (0.03)	1	2	0	0	0	0
Culex poicilipes	57 (0.50)	34	1	22	0	0	0
Culex quinquefasciatus	9 (0.08)	1	0	0	8	0	0
Culex tritaeniorhynchus	4467 (38.98)	319	750	810	56	1397	1135
Culex univittatus/neavei	264 (2.30)	57	86	83	38	0	0
Mansonia uniformis	262 (2.29)	172	38	50	1	0	1
Uranotaenia sp.	4 (0.03)	4	0	0	0	0	0
	11459	2144	1658	3251	234	2092	2080

($F = 25.89$, df = 1, $P < 0.001$). For each site, 36 trap nights were carried out (i.e., 16 with NT and 20 with LT). Interactions between sites and traps were highlighted ($F = 9.25$, df = 5, $P < 0.001$), suggesting that the difference between traps was not observed in all sites. For example, in the site of Bekoratsaka, we showed that the LT captured more mosquitoes than NT (Fig. 2).

For *An. coustani*, no difference between traps was observed neither in the Mampikony district ($F = 2.68$, df = 1, $P = 0.11$) nor in the Toliary II district ($F = 0.37$, df = 1, $P = 0.55$). For *An. squamosus/cydippis*, in the Mampikony district, significant difference between traps was observed ($F = 5.21$, df = 1, $P < 0.05$), i.e., the LT captured more than the NT. In the Toliary II district, this difference was not observed ($F = 1.22$, df = 1, $P = 0.29$). But, for *Cx. antennatus*, significant difference was observed in the Mampikony district ($F = 7.87$, df = 1, $P < 0.01$), i.e., the NT was more efficient that the LT. This difference was not seen in the Toliary II district ($F = 0.46$, df = 1, $P = 0.51$). For *Cx. tritaeniorhynchus*, significant difference was seen either for the Mampikony district ($F = 6.36$, df = 1, $P < 0.05$) or for the Toliary II district ($F = 4.98$, df = 1, $P < 0.05$), i.e., the NT was more efficient than the LT.

Host Preference. For the species that the NT was more efficient than the LT, the attractiveness of the bait used in the NT was tested (i.e., the comparison was between zebu and sheep plus goat). This case concerned *Cx. antennatus* in Mampikony and *Cx. tritaeniorhynchus* for the two districts. For *Cx. antennatus*, no difference between baits was observed ($F = 0.41$, df = 1, $P = 0.53$). But, for *Cx. tritaeniorhynchus*, significant difference was observed in both Mampikony ($F = 4.92$, df = 1, $P < 0.05$) and Toliary II ($F = 4.98$, df = 1, $P < 0.05$), i.e., the zebu was more attractive than the sheep or goat.

Temporal Pattern in Species Distribution. According to season, difference between the abundance of mosquitoes and seasons was observed ($F = 29.45$, df = 1, $P < 0.001$). Significant difference was observed for both the Mampikony district ($F = 17.89$, df = 1, $P < 0.001$) and the Toliary II district ($F = 6.83$, df = 1, $P < 0.01$; Fig. 3). We have shown an interaction between site and seasons ($F = 3.25$, df = 5, $P < 0.01$), concluding that, the abundance of mosquito varies from site-to-site according to season.

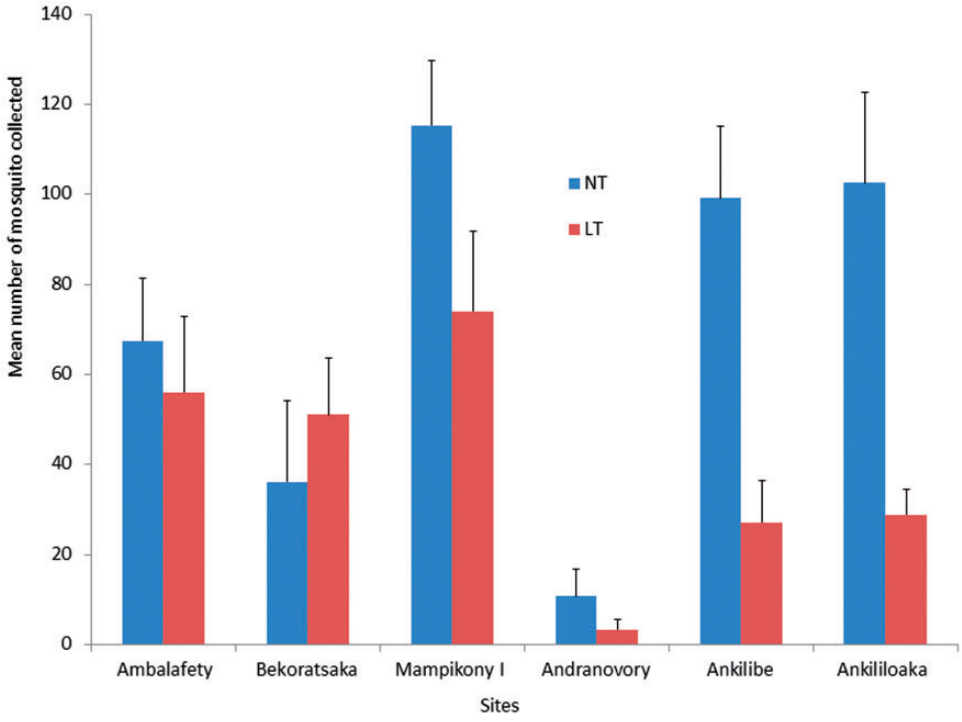


Fig. 2. Mean number of total mosquitoes collected in animal-baited net traps (NTs) or CDC light traps without CO_2 (LTs) in two districts of Madagascar. Statistical analyses showed a significant difference among the mean number of mosquitoes per site caught by the two traps ($F = 25.89$, df = 1, $P < 0.001$). For each site, 36 trap nights were carried out (i.e., 16 with NT and 20 with LT). Interactions between sites and traps were highlighted ($F = 9.25$, df = 5, $P < 0.001$), only in the site of Bekoratsaka that the LT captured more mosquitoes than the NT.

Diversity Analysis. The Jaccard index between the two districts indicated 49% of common species between the two districts (J = 0.49). The value between the two districts was influenced by the repartition of the species belonging to the *Aedes* genus. Indeed, with the six species of *Aedes*, no species was common for the two districts. Eleven species were captured only at Mampikony, whereas five were unique to Toliary II.

The Simpson index value showed that the probability to collect the same species in the Toliary district (D = 0.41) was higher than in the Mampikony district (D = 0.14). The Shannon's equitability indexes value were $E_H = 0.69$ for the Mampikony district and $E_H = 0.34$ for the Toliary II district. It means that more evenness was observed in the Mampikony district than in the Toliary II district. These biodiversity indexes were confirmed by the difference between the number of mosquitoes according to species ($F = 6.22$, df = 33, $P < 0.001$). This difference between the numbers of mosquitoes by species was also observed for each district, $F = 7.81$, $P < 0.001$, df = 30, for the Mampikony district, and $F = 1.70$, $P < 0.05$, df = 23, for the Toliary II district.

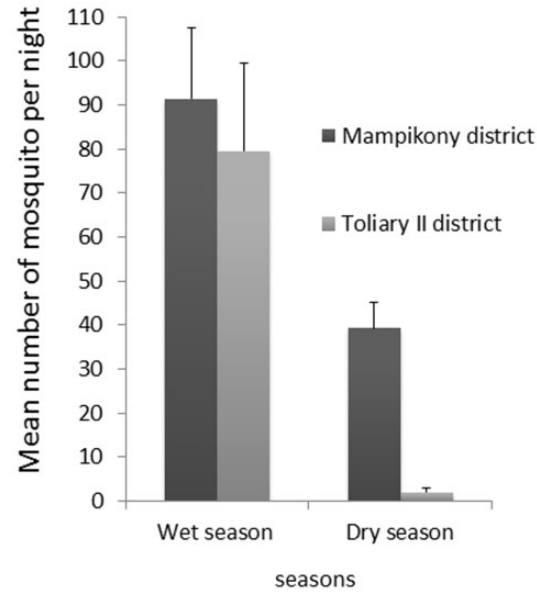


Fig. 3. Mean number of mosquitoes per night, per district, and per season. Difference between the mean of mosquitoes and seasons was observed ($F = 29.45$, df = 1, $P < 0.001$). This difference was observed for both the Mampikony district ($F = 17.89$, df = 1, $P < 0.001$) and the Toliary II district ($F = 6.83$, df = 1, $P < 0.01$).

Discussion

The abundance of mosquito and the species richness were different in the two districts. Temporal and spatial

dynamics in mosquito abundances are influenced by environmental stress (Ribeiro et al. 2012), environmental variables and regional climate such as precipitation, temperature, humidity, land use/cover, and hydric soils located in a 1-km radius (Buckner et al. 2011, Ganser and Wisely 2013). Other factors could influence the abundance and diversity of mosquitoes, such as the distance between the breeding site and trap (Diallo et al. 2011) and the trap location (L'Ambert et al. 2012). In the Mampikony district, the presence of a permanent lake and river near the village could explain the higher abundance and important diversity of mosquito in each site. In Toliary II, the breeding sites are further, and rivers or lakes are drained during the dry season. Moreover, the Toliary II district is influenced by coastline, explaining also the difference in composition and in abundance of mosquito in the two districts. In addition, our study was the first entomological investigation in the district of Mampikony, and for the Toliary II district, the last entomological investigation was carried out in 1989 in the town of Toliary and around (Fontenille 1989). While this previous study described the dominance of *Ae. durbanensis* and *Aedes fryeri* (Theobald) species and the absence of *Cx. antennatus* species (Fontenille 1989), our study has shown the dominance of *Cx. tritaeniorhynchus* and *Ae. durbanensis*, the presence of *Cx. Antennatus*, and the absence of *Ae. fryeri*. These differences may be explained by the sampling tools used but also by the short-term sampling of this study.

The NT trapped more mosquitoes than LT in abundance and in composition. This study supported that chosen traps during study affect the species abundance and composition estimates (Brown et al. 2008). But, the significant interaction between trap and site effect showed that these traps were complementary. In NT, mosquitoes were attracted by the cue emitted by the bait, i.e., odor, carbon dioxide, warmth, and moisture. Despite the abundant of zebus in the two districts, no specific host preference was observed, except for *Cx. tritaeniorhynchus* with zebu. In the absence of zebu, goat, or sheep, mosquitoes in the area could opportunistic-feed on human or other host which are more abundant or convenient bloodmeal sources (Chaves et al. 2010, de Carvalho et al. 2014). Moreover, most of these species were already caught in Madagascar by human landing catch (Andrianaivolambo et al. 2010, Ratovonjato et al. 2014). By contrast, species that feed on wild host could not be estimated with our study, such as the genus *Uranotaenia* (Lynch Arribàzaga), which principally feed on amphibians and reptiles (da Cunha Ramos and Brunhes 2004).

Regarding to RVFV transmission, 24 Malagasy mosquito species were involved in RVFV transmission (Balenghien et al. 2013, Tantely et al. 2015). Eighteen of them were collected during this study: *Aedes circumluteolus* (Theobald; Weinbren et al. 1957), *Ae. fowleri* (Turell et al. 1988), *Ae. durbanensis, An. pharoensis* (Linthicum et al. 1985), *An. arabiensis* (Fontenille et al. 1998), *An. gambiae* (Moutailler et al. 2008), *An. pauliani/radama* (Clerc et al. 1981), *An. coustani, An. squamosus/cydippis*, *Cx. antennatus* (Ratovonjato et al. 2011), *Culex neavei* (Theobald), (McIntosh and Jupp 1981), *Culex pipiens* (Linné; Hoogstraal et al. 1979), *Culex bitaeniorhynchus* (Giles), *Culex poicilipes* (Theobald), *Culex quinquefasciatus* (Say), *Culex. univittatus* (Theobald), *Mansonia uniformis* (Theobald; Sang et al. 2010), and *Cx. tritaeniorhynchus* (Jupp et al. 2002). Among these species, *Cx. tritaeniorhynchus*, *Cx. antennatus*, *An. squamosus/cydippis, Ae. durbanensis*, *An. coustani*, and *An. pauliani/radama* were the most abundant species during this study, representing 82.33% of the catching. RVFV was isolated from *Cx. antennatus, An. squamosus/cydippis*, and *An. coustani* in Madagascar in 2008 and 2009 (Ratovonjato et al. 2011). We showed that these three species were more abundant in the Mampikony district than in the Toliary II district: this difference in abundance of these potential vectors could explain the prevalence of RVF infection in ruminants previously described in the two areas (Jeanmaire et al. 2011).

We have also captured other mosquito species involved in *Plasmodium* and West Nile virus transmission. According to malaria vectors in these districts, *An. merus*, *An. mascarensis*, and *An. arabiensis* species were present in the Toliary II district. The presence of *An. funestus* was detected in Toliary in 1989 (Fontenille 1989). *An. mascarensis* , *An. arabiensis*, *An. gambiae ss*, and *An. funestus* species were present in the Mampikony district. We also captured seven species out of the 14 Malagasy mosquito species that have already been involved in West Nile virus transmission (Tantely et al. unpublished): *Cx. quinquefasciatus*, *Cx. tritaeniorhynchus*, *Cx. univittatus*, *Cx. neavei*, *Culex decens* (Theobald; Hubálek and Halouzka 1999), *Culex pipiens* (L.), and *Cx. antennatus* (Zeller and Schuffenecker 2004). This virus was isolated from *Cx. quinquefasciatus*, *Cx. tritaeniorhynchus, Cx. decens*, and *Cx. antennatus* in Madagascar in 1985 and 1986 (Fontenille 1989). These species were present in the two districts, but their exact role in the transmission cycle of West Nile virus remains unknown.

To conclude, Madagascar is in an inter-epidemic period transmission of RVF. The presence of mosquito species involved in RVFV transmission either in Madagascar or in other countries was mentioned in the areas of sampling. Despite the catching of these species by using domestic animals as attractants, they also have opportunist behavior, and they are able to bite humans (Tantely et al. 2015). These findings are important to understand the actors and their role of RVFV and other pathogens transmission in Madagascar.

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